

Machine Learning for Asset Return Prediction

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Abstract

This study tests whether lagged asset characteristics and macroeconomic information can forecast next-month excess returns across 50 anonymized assets spanning four asset classes and 12 countries from 2000 through 2024. I compare OLS, Elastic Net, Random Forest, Gradient Boosting, and forecast ensembles using strictly time-ordered expanding and ten-year rolling estimation schemes.

The strongest specification is a rolling Elastic Net combining asset characteristics with four global macroeconomic variables. It achieves an out-of-sample R^2 of 0.467% relative to zero and 0.492% relative to a real-time trailing-mean benchmark. Country-level macro variables, characteristic-macro interactions, and a wider high-dimensional macro panel do not improve performance.

Translating the preferred forecasts into monthly portfolio positions produces a net Sharpe ratio of 0.439 at 10 basis points per dollar of turnover, compared with 0.433 for Equal Weight, while operating at substantially lower volatility and drawdown. The improvement is modest, sensitive to transaction costs, and strongly related to Risk Parity exposure.

Overall, the evidence favors regularization, recent information, and disciplined out-of-sample validation over greater model complexity.

1. Introduction

Predicting expected returns is a central problem in empirical asset pricing and practical portfolio allocation. Even modest improvements in forecasts can be economically useful if they help distinguish relatively attractive assets, adapt exposures to changing economic conditions, or improve portfolio risk-adjusted performance. At the same time, realized returns are extremely noisy, and relationships that appear strong in sample often fail when evaluated on new data.

The relevant question is therefore not simply whether characteristics or macroeconomic variables are historically correlated with returns, but whether information available to an investor at a given date can forecast returns that occur afterward.

This study examines monthly excess-return predictability in a panel of 50 anonymized assets spanning four asset classes and 12 countries from January 2000 through December 2024. The analysis addresses three questions:

- How much of next month's excess return can be forecast from lagged asset characteristics?
- Does global or country-specific macroeconomic information add predictive content beyond those characteristics?
- Can the resulting forecasts improve on Equal Weight, Risk Parity, and Time-Series Momentum after turnover and transaction costs?

The empirical design is motivated by two complementary strands of the asset-pricing literature. Gu, Kelly, and Xiu (2020) show that machine-learning methods can uncover nonlinearities and interactions in large predictor sets. Welch and Goyal (2008), however, demonstrate that many apparently successful macroeconomic return predictors perform poorly once evaluated out of sample. Together, these findings motivate a design centered on chronological validation, regularization, real-time information availability, and explicit benchmark comparisons.

The data create several practical challenges. Return volatility varies substantially across asset classes, asset characteristics differ in scale and persistence, and the macroeconomic panel contains persistent and correlated variables. To prevent high-volatility assets from dominating the pooled forecasting objective, I predict volatility-scaled excess returns rather than raw returns.

Rather than committing to a single model, I compare OLS, Elastic Net, Random Forest, Gradient Boosting, and an ensemble under both expanding and rolling estimation windows. I then test richer characteristic-macro interactions and a substantially wider macroeconomic panel before translating the strongest forecasting specification into an investable portfolio.

The central result is that greater complexity does not produce better forecasts. The strongest results come from a relatively parsimonious rolling Elastic Net using asset characteristics and four global macro variables. The resulting portfolio provides a modest improvement in risk-adjusted performance at low transaction costs, but its behavior remains closely related to a simple Risk Parity allocation.

2. Data and Feature Construction

2.1 Asset and Macroeconomic Data

The dataset is a balanced monthly panel from January 2000 through December 2024. It contains 50 assets observed over 300 months, for a total of 15,000 asset-month observations. Returns are monthly excess returns.

The universe spans four anonymized asset classes. Classes B, C, and D have economically meaningful assignments to 12 anonymized countries. Class A, which contains 26 of the 50 assets, has no natural country assignment.

Each asset is accompanied by five anonymized asset-level characteristics (x_1 through x_5). The macroeconomic data include four global variables (x_6 through x_9), seven curated country variables (x_{10} through x_{16}), and a wider set of 148 additional country-level variables (x_{17} through x_{164}).

Exploratory analysis shows substantial differences in risk across asset classes. Class A represents approximately 52% of the assets but contributes roughly 87% of total squared raw returns. This motivates the use of a volatility-scaled target in the pooled forecasting exercise.

The supplied characteristic x_5 closely tracks trailing 36-month realized volatility and is used as the lagged volatility estimate. The macroeconomic variables are generally persistent, with the four global variables exhibiting high first-order autocorrelation.

The curated country variable x_{11} stops updating before the end of the sample for three countries. Rather than forward-fill stale observations or use an imputation method that could introduce unavailable information, x_{11} is excluded.

2.2 Prediction Target

The forecasting target is volatility-scaled monthly excess return,

$$y_{i,t} = \frac{r_{i,t}}{\hat{\sigma}_{i,t-1}}, \quad (1)$$

where $r_{i,t}$ is the month- t excess return and $\hat{\sigma}_{i,t-1}$ is the lagged volatility estimate.

This transformation prevents the highest-volatility assets from dominating a pooled squared-error objective.

A forecast of the scaled target maps back to raw excess return as

$$\hat{r}_{i,t} = \hat{y}_{i,t} \hat{\sigma}_{i,t-1}. \quad (2)$$

2.3 Predictor Construction

All predictors are aligned so that only information observable before the month- t return is used.

The five asset characteristics are converted to cross-sectional percentile ranks within asset class and month and rescaled to the interval $[-1, 1]$. Three class indicators are added, with Class A omitted.

The four global macroeconomic variables are lagged by one month.

For country-level data, the six complete curated variables x_{10} and x_{12} through x_{16} are lagged by one month, ranked cross-sectionally across the 12 countries at each date, and rescaled to $[-1, 1]$.

Assets in Classes B, C, and D receive the lagged rank associated with their country. Because Class A has no meaningful country assignment, its country-specific predictors are set to zero, the neutral midpoint of the rank transformation.

Table 1 summarizes the three core information sets.

Table 1: Core predictor sets

Information set	Predictors	Count
CHAR	Five transformed asset characteristics and three asset-class indicators	8
GLOBAL	CHAR plus lagged global variables x_6 – x_9	12
COUNTRY	GLOBAL plus lagged country ranks x_{10} and x_{12} – x_{16}	18

The nested structure allows model complexity and information content to be evaluated separately. Comparisons across models within an information set test whether greater functional flexibility improves forecasts. Comparisons across CHAR, GLOBAL, and COUNTRY within the same model test whether macroeconomic variables provide incremental information.

3. Empirical Design

3.1 Forecasting Models

Four base forecasting models are estimated for each core information set.

OLS provides an unregularized linear benchmark.

Elastic Net combines ridge and lasso penalties. The penalty parameter is selected from $\{0.001, 0.01, 0.1, 0.3, 1\}$ and the lasso mixing parameter from $\{0.1, 0.5, 0.9, 1\}$.

Random Forest uses 300 trees. Maximum tree depth is selected from $\{3, 6\}$, minimum leaf size from $\{10, 30\}$, and the fraction of predictors considered at each split from $\{0.5, 1.0\}$.

Gradient Boosting selects between 100 and 300 trees, learning rates of 0.03 or 0.10, tree depths of 1 or 2, and a minimum leaf size of 20.

Finally, an **ensemble** takes the equal-weight average of the Elastic Net, Random Forest, and Gradient Boosting forecasts.

3.2 Time-Ordered Validation

The final out-of-sample evaluation period is January 2015 through December 2024. Models are refit annually and generate monthly forecasts during each test year.

Two estimation schemes are considered.

Under the **expanding-window** design, the initial training period ends in 2009, the validation period spans 2010–2014, and 2015 is the first test year. The historical estimation window then expands through time.

Under the **rolling-window** design, the model uses the most recent 10 years of training data, a subsequent 5-year validation block, and a 1-year test block.

All splits are made by date rather than by asset-month observation, so every asset in a given month remains in the same sample. Random cross-validation is never used.

For models requiring hyperparameter selection, candidate specifications are evaluated on the historical validation block using mean squared forecast error. After hyperparameters are selected, the model is refit using all historical observations permitted by the corresponding expanding or rolling scheme. The test year is never used for estimation or tuning.

Predictor standardization and other fitted transformations are computed from historical estimation data only.

3.3 Forecast Evaluation

Forecasts are evaluated against two real-time benchmarks:

- zero expected scaled excess return; and
- the pooled expanding historical mean of scaled excess returns through the previous month.

For benchmark forecast $\hat{y}_{i,t}^b$,

$$R_{\text{OOS}}^2 = 1 - \frac{\sum_{i,t} (y_{i,t} - \hat{y}_{i,t})^2}{\sum_{i,t} (y_{i,t} - \hat{y}_{i,t}^b)^2}. \quad (3)$$

A positive value indicates lower out-of-sample forecast error than the benchmark.

3.4 Additional Specifications

Two extensions test whether the core design is too restrictive.

The first adds all 20 interactions between the five asset characteristics and four global macro variables to the Elastic Net. Combined with the original characteristics, class indicators, and global variables, this produces 32 predictors.

The second uses the wider country-macro panel. Missing, exact-duplicate, and constant series are removed, eliminating 18 of 148 variables and leaving 130 candidate predictors.

Because directly fitting a flexible nonlinear model to all 130 variables would create a high-dimensional search problem, I use a two-stage rolling procedure. The 10-year historical training block is divided into a 7-year screening-estimation period and a 3-year screening-validation period. Elastic Net selects the penalty and identifies extended macro variables with nonzero coefficients. It is then refit on the full 10-year training block, and only surviving extended variables are retained.

The five characteristics, three class indicators, and four global variables are always retained. Gradient Boosting is then tuned using the separate 5-year validation period, refit on the most recent 10 years, and used to forecast the following test year.

4. Out-of-Sample Forecasting Results

4.1 Core Forecasting Performance

Table 2 reports the complete core forecasting comparison.

Under expanding estimation, no characteristic-only model produces a positive R_{OOS}^2 relative to zero.

The results improve when estimation is restricted to the most recent 10 years. With the CHAR information set, rolling Elastic Net, Random Forest, and the ensemble all achieve small positive performance relative to both benchmarks.

The largest improvement occurs when global macroeconomic variables are added to the rolling Elastic Net. Its exact R_{OOS}^2 rises from 0.142% under CHAR to 0.467% under GLOBAL relative to zero. Relative to the trailing-mean benchmark, GLOBAL Elastic Net achieves 0.492%.

Adding country-specific variables yields essentially no further improvement.

Table 2: Core out-of-sample forecasting performance

Estimation	Model	CHAR	GLOBAL	COUNTRY
<i>Panel A: R^2_{OOS} versus zero benchmark</i>				
Expanding	OLS	-0.32%	-1.61%	-1.61%
Expanding	Elastic Net	-0.04%	-0.03%	-0.03%
Expanding	Random Forest	-0.22%	-4.52%	-6.02%
Expanding	Gradient Boosting	-0.20%	-2.64%	-2.64%
Expanding	Ensemble	-0.10%	-1.79%	-2.19%
Rolling	OLS	-0.07%	-0.72%	-0.75%
Rolling	Elastic Net	0.14%	0.47%	0.46%
Rolling	Random Forest	0.12%	-2.77%	-2.77%
Rolling	Gradient Boosting	-0.04%	-1.79%	-1.07%
Rolling	Ensemble	0.15%	-0.70%	-0.50%
<i>Panel B: R^2_{OOS} versus trailing-mean benchmark</i>				
Expanding	OLS	-0.30%	-1.58%	-1.58%
Expanding	Elastic Net	-0.01%	-0.01%	-0.01%
Expanding	Random Forest	-0.20%	-4.50%	-5.99%
Expanding	Gradient Boosting	-0.17%	-2.62%	-2.62%
Expanding	Ensemble	-0.08%	-1.77%	-2.16%
Rolling	OLS	-0.05%	-0.69%	-0.72%
Rolling	Elastic Net	0.17%	0.49%	0.49%
Rolling	Random Forest	0.14%	-2.75%	-2.74%
Rolling	Gradient Boosting	-0.01%	-1.76%	-1.04%
Rolling	Ensemble	0.17%	-0.68%	-0.47%

Note: Entries are percentage points and are rounded to two decimal places. Exact values for the preferred specification are reported in the text.

Although the magnitudes are small, monthly realized returns contain substantial idiosyncratic noise. The relevant comparison is therefore whether a model consistently reduces forecast error relative to real-time benchmarks, not whether it explains a large share of total monthly return variation.

The macroeconomic results are also model-dependent. Global information improves the regularized Elastic Net but substantially worsens Random Forest, Gradient Boosting, OLS, and the ensemble. Simply expanding the information set therefore does not necessarily improve forecasting performance.

4.2 Heterogeneity Across Asset Classes

Table 3 evaluates the same pooled rolling GLOBAL Elastic Net within each asset class.

Table 3: Rolling GLOBAL Elastic Net performance by asset class

Class	R^2_{OOS} vs. zero	R^2_{OOS} vs. trailing mean
A	0.675%	0.467%
B	-0.035%	1.899%
C	1.800%	0.490%
D	-1.368%	-0.841%

Performance is heterogeneous. The preferred model is strongest in Class C and also positive in Class A relative to both benchmarks. Class B is nearly flat relative to zero but substantially better than its historical-mean benchmark, while Class D is negative relative to both.

These results come from the same pooled model rather than four separately estimated class-specific models.

4.3 Interactions and Extended Macroeconomic Information

The richer specifications do not improve the core result.

Adding all characteristic-global-macro interactions produces a rolling R^2_{OOS} of 0.459%, compared with 0.467% for the simpler GLOBAL Elastic Net. Under expanding estimation, the interaction model falls to -1.308%.

The interaction coefficients are also unstable: no interaction is selected in more than one of the 10 rolling annual estimation windows, and most are never selected.

The wider country-macro experiment reaches a similar conclusion. After cleaning, 130 extended macro predictors remain. The screened Gradient Boosting specification produces a rolling R^2_{OOS} of -1.794%, essentially unchanged from the core GLOBAL Gradient Boosting result of -1.788%. No extended macro variable is selected in more than two of the 10 annual windows.

Taken together, the extensions provide little evidence that additional functional flexibility or substantially greater predictor dimensionality uncovers a stable forecasting signal.

5. Portfolio Evaluation

5.1 From Forecasts to Portfolio Weights

The rolling GLOBAL Elastic Net is used for the portfolio analysis because it delivers the strongest overall forecasting performance among the core specifications.

Because the model forecasts volatility-scaled return, I define the portfolio score as

$$s_{i,t} = \frac{\hat{y}_{i,t}}{\hat{\sigma}_{i,t-1}} = \frac{\hat{r}_{i,t}}{\hat{\sigma}_{i,t-1}^2}. \quad (4)$$

The second volatility adjustment converts the forecast into a return-per-variance score, favoring assets with high predicted return and low risk.

Scores are normalized to unit gross exposure,

$$w_{i,t} = \frac{s_{i,t}}{\sum_j |s_{j,t}|}, \quad \sum_i |w_{i,t}| = 1. \quad (5)$$

Positive scores generate long positions and negative scores generate short positions. No separate net-exposure constraint is imposed.

Weights are formed before the month- t realized return and rebalanced monthly.

The forecast strategy is compared with three unit-gross benchmarks:

- **Equal Weight:** equal positions across all 50 assets;
- **Risk Parity:** weights proportional to inverse trailing 36-month volatility;
- **Time-Series Momentum:** the sign of trailing 12-month return divided by trailing volatility.

Turnover is computed from changes in target weights and applied consistently across strategies. The primary transaction-cost assumption is 10 basis points per dollar of turnover, with sensitivity analyses at 25 and 50 basis points.

Because the preferred model is selected using forecasting performance over the same 2015–2024 period used for the portfolio comparison, the portfolio exercise should be viewed as an economic evaluation of the preferred

forecasting specification rather than as a completely independent second holdout sample. The individual forecasts and portfolio weights themselves remain genuinely out of sample.

5.2 Portfolio Performance

Table 4 reports performance from 2015 through 2024.

Table 4: Out-of-sample portfolio performance, 2015–2024

Strategy	Gross ret.	Vol.	Gross SR	Net ret.	Net SR	Max DD	Turnover
Equal Weight	3.80%	8.78%	0.433	3.80%	0.433	-17.55%	0.00%
Risk Parity	1.61%	5.38%	0.299	1.59%	0.294	-10.70%	2.11%
Time-Series Momentum	0.34%	4.02%	0.085	0.04%	0.010	-13.96%	25.47%
Forecast Elastic Net	2.36%	5.13%	0.459	2.25%	0.439	-10.70%	8.93%

Note: Net performance assumes transaction costs of 10 basis points per dollar of turnover. Returns and volatility are annualized.

At 10 basis points, the forecast portfolio earns a net annualized excess return of 2.25% with annualized volatility of 5.13%, producing a net Sharpe ratio of 0.439.

Equal Weight earns a higher annualized return of 3.80%, but with materially higher volatility of 8.78% and a Sharpe ratio of 0.433. The forecast portfolio therefore produces slightly more return per unit of risk, although not a higher absolute return.

Risk Parity produces similar volatility but a lower net Sharpe ratio of 0.294. Time-Series Momentum performs poorly during the evaluation period, with a net Sharpe ratio of 0.010.

The forecast strategy's maximum drawdown is approximately -10.70%, compared with -17.55% for Equal Weight.

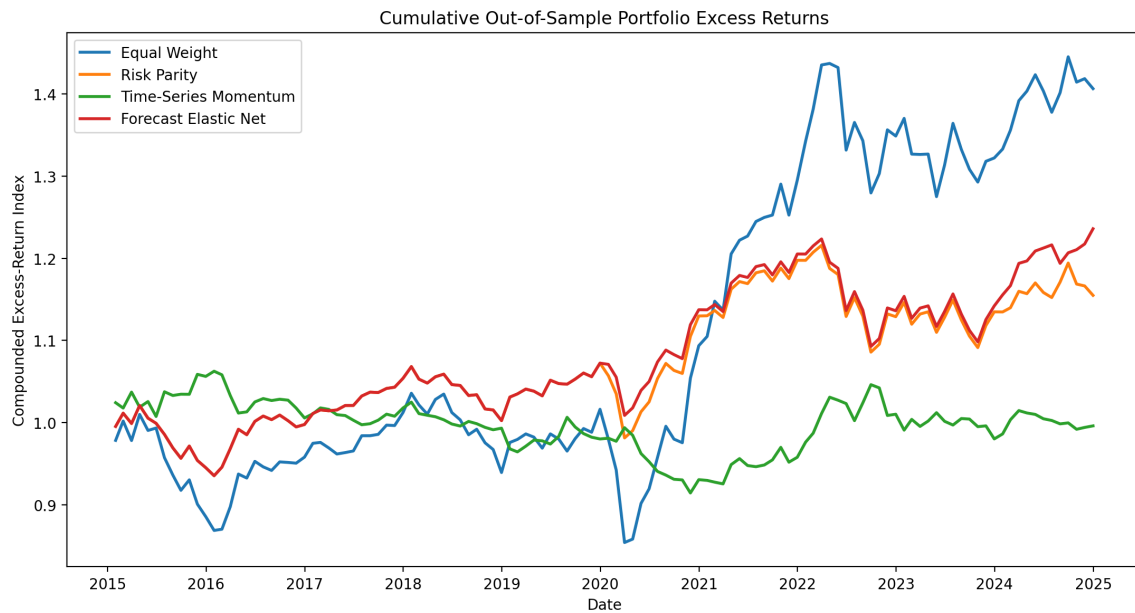


Figure 1: Cumulative out-of-sample net portfolio excess returns, 2015–2024. Net returns incorporate transaction costs of 10 basis points per dollar of turnover.

The portfolio advantage is economically modest and sensitive to trading costs. Increasing costs from 10 to 25 basis points reduces its net Sharpe ratio from 0.439 to 0.409. At 50 basis points, the net Sharpe falls to 0.358. Corresponding net annualized returns decline from 2.25% to 2.09% and 1.83%.

The forecast portfolio therefore does not dominate Equal Weight. Its advantage is instead a modest improvement in

risk-adjusted performance and drawdown at low implementation costs.

5.3 Benchmark Exposure

The portfolio's lower volatility raises an important question: how much of its behavior simply reflects exposure to standard allocation rules?

The forecast portfolio's monthly net returns have a correlation of 0.939 with Risk Parity, 0.866 with Equal Weight, and -0.361 with Time-Series Momentum.

A regression of forecast-portfolio returns on the three benchmarks produces a Risk Parity loading of 0.851, compared with 0.022 on Equal Weight and -0.037 on Time-Series Momentum. Together, the benchmarks explain 88.27% of monthly forecast-portfolio return variation.

A decomposition of average annual return attributes approximately 1.35 percentage points of the forecast portfolio's 2.25% annual net return to Risk Parity exposure, 0.08 percentage points to Equal Weight exposure, approximately zero to Time-Series Momentum, and 0.82 percentage points to the regression intercept.

Because no standard error or significance test is attached to the intercept, the 0.82% figure should be interpreted as descriptive rather than as statistically established alpha.

The forecast strategy therefore behaves primarily like a dynamically modified Risk Parity portfolio with a smaller return component not captured by the three benchmark rules.

6. Interpretation

6.1 Why Does the Rolling Window Perform Better?

The difference between expanding and rolling estimation is one of the clearest patterns in the results.

One possible explanation is that return-predictive relationships are not fully stable over the full sample. Financial markets evolve with monetary regimes, regulation, technology, investor participation, and the composition of priced risks. If relationships from the early 2000s are less relevant in the 2020s, retaining all historical observations may dilute more recent information.

The rolling window therefore trades sample size for relevance. It discards older observations and increases estimation uncertainty, but implicitly places greater weight on relationships observed during the most recent decade.

The results are consistent with the benefit of relevance outweighing the cost of the smaller sample in this dataset. They do not, however, establish structural change as the cause; some of the difference could reflect sampling variation during the particular 2015–2024 evaluation period.

6.2 Why Does Elastic Net Benefit from Global Macro Information?

The global macro results suggest that aggregate economic conditions contain incremental information about expected returns, but that this information is weak relative to the noise in the predictor set.

Elastic Net is well suited to such an environment because it shrinks coefficient estimates and can set weak predictors to zero. This limits the influence of unstable relationships.

The contrast across models is informative. Once global macro variables are introduced, Elastic Net improves substantially while OLS and the more flexible tree-based models deteriorate.

Random Forest and Gradient Boosting can represent nonlinearities and interactions, but flexibility is valuable only if the sample contains enough stable information to estimate those relationships. Here, additional flexibility appears to increase estimation noise faster than it increases useful signal.

This should not be interpreted as evidence that nonlinear relationships are unimportant in asset pricing generally.

A narrower conclusion is supported: with monthly observations, 50 assets, a relatively short time dimension, and weak return signal, these nonlinear models do not generalize successfully in this application.

6.3 Global Versus Country-Specific Information

Global macro variables improve the preferred rolling Elastic Net, while the curated country variables provide almost no additional gain once characteristics and global information are included.

Common global financial conditions may simply be more important for this asset universe than country-specific variation. The country variables may also be persistent or redundant with the global state.

In addition, Class A accounts for more than half of the asset universe but has no natural country assignment, limiting the role that country-specific information can play consistently across the full panel.

The result therefore should not be interpreted as evidence that country fundamentals are economically irrelevant. It shows more narrowly that the available country-level variables do not provide stable incremental next-month forecasting value under the transformations, timing conventions, and models considered here.

6.4 Why Does Additional Complexity Not Help?

The interaction and extended-macro experiments test whether the core design is simply too restrictive.

The characteristic-macro interaction model allows the effect of an asset characteristic to depend on the global macroeconomic state. Although this is economically plausible, the model does not improve out-of-sample performance, and selected interaction coefficients are unstable across forecasting windows.

The extended country-macro experiment reaches the same conclusion from a different direction. Even after a regularized screening stage, the much larger macro panel fails to improve the nonlinear model and selected variables vary considerably through time.

Together, these experiments suggest that neither greater functional flexibility nor substantially greater predictor dimensionality uncovers a stable additional signal. In this application, parsimony is more valuable than additional complexity.

7. Limitations and Further Research

Several limitations should shape the interpretation of the results.

First, although each individual test-year forecast is genuinely out of sample, the rolling GLOBAL Elastic Net is selected after comparing models over the 2015–2024 evaluation period. The portfolio analysis therefore does not constitute a completely independent second holdout test. A stricter design could reserve a final block of years exclusively for portfolio evaluation after all forecasting decisions are fixed.

Second, the improvements themselves are small, and I do not conduct formal statistical tests of whether the differences in R^2_{OOS} or Sharpe ratios are statistically distinguishable from the benchmarks. The results should therefore be viewed as descriptive evidence of improvement rather than definitive statistical superiority.

Third, transaction costs are calculated from changes in target weights because the available excess-return data do not permit exact reconstruction of post-return portfolio weights. The convention is applied consistently across strategies but remains an approximation to realized turnover.

Fourth, the forecasting models are pooled across asset classes. Class indicators allow for broad differences and performance is evaluated separately by class, but class-specific models are not estimated. Given the heterogeneity in class-level forecasting performance, a larger sample could test whether characteristic and macro relationships differ systematically across asset classes.

The analysis also identifies the incremental value of broad predictor groups rather than attributing the preferred model's performance to individual characteristics or global macro variables. Studying the stability, sign, and

magnitude of Elastic Net coefficients across forecasting windows would provide a more detailed view of the underlying signals.

Finally, additional macroeconomic placebo tests, including deliberately misaligning persistent macro variables by several alternative lags, could help quantify how much apparent predictive power may arise from persistence and chance.

8. Conclusion

This study evaluates whether asset characteristics and macroeconomic information can forecast monthly excess returns across a diversified panel of 50 assets and whether those forecasts are economically useful for portfolio allocation.

Three conclusions emerge.

First, asset characteristics contain modest out-of-sample predictive information, but the signal is stronger when estimation focuses on the most recent 10 years rather than all available history. The result is consistent with time variation in return-predictive relationships, although it does not prove that structural change is the cause.

Second, macroeconomic information is useful selectively rather than universally. Four global macro variables materially improve the rolling Elastic Net, raising R^2_{OOS} relative to zero from 0.142% to 0.467%. Country-specific variables provide almost no additional gain. Neither explicit characteristic-macro interactions nor the wider country-macro panel improves forecasting performance. More flexible Random Forest and Gradient Boosting models also deteriorate once macro variables are introduced.

Third, the forecasting improvement has economic value, but the magnitude is limited. The forecast-based portfolio achieves lower volatility and drawdown than Equal Weight and a slightly higher Sharpe ratio when trading costs are low. It does not earn a higher raw return, and its risk-adjusted advantage declines as costs rise. Most of its return variation is also closely related to Risk Parity.

For a portfolio manager, the implication is therefore not that machine learning replaces simple allocation rules. The evidence instead supports using a regularized forecasting model as a modest adjustment to a disciplined risk-allocation process.

More broadly, the project illustrates a central difficulty of empirical return prediction: statistical flexibility is abundant, but stable signal is scarce. In this sample, careful time ordering, regularization, and model restraint matter more than simply adding predictors or complexity.

References

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